are more likely to exploit loopholes (compared to outright defiance) when the cost of non-compliance is relatively high, such as in an upward relationship (Bridgers et al., 2023).

In the evaluation task, the scenario includes the relationship between the protagonist and the person that issues the directive, the uttered directive, as well as the protagonist's behavior in response to the directive: either compliance, non-compliance, or exploiting a loophole (see Figure 1, left). In the generation task, the scenario includes the same background information about the relationship and the directive uttered by another person, as well as the protagonist's intentions. Models and humans are prompted to come up with a loophole response (see Figure 1, right). A sample of these scenarios are shown in Table 1 in Appendix B.

2.2 Collecting Human Responses

For the evaluation task, we used data from Bridgers et al. (2023). They recruited 180 US adults with above a 95% approval rating online via Prolific (Peer et al., 2017) for the evaluation task. Participants (M_{age}: 35.06; 50% female; 70% White, 9% Hispanic or Latinx, 7% Black/ African American, 6% Mixed, 5% Asian, 3% Other) were U.S. residents fluent in English and from diverse regional and educational backgrounds. An additional 7 participants were recruited but excluded from analysis due to failing an attention check. The survey took approximately 17 minutes to complete, and compensation was $4.04. Participants saw 12 scenarios: 4 ending in compliance, 4 in loopholes, and 4 in non-compliance (see Appendix A for more information on the experimental setup).

For the generation task, we recruited 52 participants on Prolific (M_{age}: 32.8, range: 18 to 59 years, 54% female, 2% non-binary) with a 95% approval rating, who lived in the U.S., and were fluent in English. The survey took approximately 9 minutes to complete, and compensation was $2.38. Participants were majority White (67%; 4% Black, 10% Hispanic or Latinx, 11% Asian, 6% multi-racial) from diverse regional and educational backgrounds. An additional 8 participants were recruited but excluded from analysis due to failure to pass an attention check. Participants were presented with a definition of loophole behavior and were then shown three examples of protagonists engaging in loopholes. Following this training, participants were asked to generate loopholes for a random subset of 12 scenarios out of the 36 scenarios we designed.

2.3 Models

We test a variety of models that have been fine-tuned to follow instructions and align with human feedback. Among these are the 3B and 11B parameter **T*k*-Instruct** models (Wang et al., 2022) and three **Flan-T5** models (base: 250M parameters; XL: 3B parameters; XXL: 11B parameters) (Chung et al., 2022). These models are all based on T5 (Raffel et al., 2019) and instruction-finetuned on a diverse collection of tasks (Wei et al., 2022). These models were accessed via Huggingface (Wolf et al., 2019). We also test three OpenAI model instances, including **InstructGPT** (`davinci-instruct-beta`), **GPT-3** (`text-davinci-003`), and **GPT-3.5** (`gpt-3.5-turbo`). InstructGPT (175B parameters) was trained using supervised fine-tuning on human demonstrations (SFT) (Ouyang et al., 2022). GPT-3 (`text-davinci-003`) was trained using reinforcement learning with reward models trained on comparisons by humans (PPO) (Brown et al., 2020). The GPT-3.5 model (`gpt-3.5-turbo`) was trained using Reinforcement Learning from Human Feedback (RLHF) on an initial model that was fine-tuned on a dialogue data supplement to the InstructGPT training data (OpenAI, 2022).[3] These models were accessed via the OpenAI API. We chose models that were representative of state-of-art systems (at the time of writing) for zero-shot, instruction-based prompting, and that have been used to test pragmatic reasoning in related works (Hu et al., 2022; Yu et al., 2023; Ruis et al., 2022).

2.3.1 Implementation Details

We designed our model assessments to be as faithful as possible to the human experiments conducted by Bridgers et al. (2023). In particular, our experimental setup makes use of *zero-shot prompting* for both the evaluation task and generation tasks, following previous works (Hu et al., 2022). This allows us to assess the abilities that emerge through the training of the models we test, and not as a result of in-context adaptation to the task. Note that for the evaluation task we simplified the format from fill-in-the-blank to question-answer (i.e.,

[3]In previous experiments we additionally tested the largest GPT-2 model (XL, 1.5B parameters; Radford et al., 2019), as a standard autoregressive language model baseline. However, its performance was so poor on both tasks that we excluded it from further analysis.

How much trouble with the protagonist get in for what they did?).

For the evaluation task, whose design is a straightforward question-answer format, we used a temperature of 0 for all models so as not to introduce unnecessary noise in the generations through a higher temperature. Since the models' output at this temperature is deterministic, we elicit a single sample per scenario. For the generation task, where more diverse model outputs were desired, we use the default temperature settings for all models, but recognize that different settings of this parameter may influence performance. Here, we elicited 5 samples per model via beam search.

2.3.2 Prompt Structure

Each prompt consisted of three parts: the task instruction, the scenario prepended with the label "Scenario:", and label for the relevant query. We use the same instructions given to the human participants.

For the evaluation task, the four valid answer options for each metric (see Section 2.4) were appended to their respective instruction lines so that they read: "How much trouble will the protagonist get into for what they did? The answer options are: no trouble, a little bit of trouble, trouble, a lot of trouble."; "How upset is the other person about what the protagonist did? The answer options are: not upset, a little bit upset, upset, very upset."; and "How funny is the protagonist's behavior to the other person? The answer options are: not funny, funny, a little bit funny, very funny.". To account for the models' sensitivity to prompt, we report the average of two versions of the evaluation prompt: one in which the valid answer options were presented in increasing order, and another in which they were presented in decreasing order. The query labels for this task corresponded to the metric for that prompt: "Trouble:", "Upset:", and "Funny:".

For the generation task, we presented the following condensed version of the instructions given to humans: "We are interested in your ability to create 'loopholes'. Here's what we mean by 'loophole behavior': People understand the actual intended meaning of a request, command, or rule, but choose to interpret things differently. Please give an example of a loophole [protagonist's name] could use." As noted before, the scenario for this task included information about protagonist's intention that is omitted in the prompt for the evaluation experiments, and the query label preceding the models' generations was "Loophole:".

2.4 Assessment Protocol

2.4.1 Behavior Evaluation Task

In the evaluation task, both human and model responses were restricted to a 4-point scale describing the amount of trouble, upset, and humor the protagonist's behavior would result in, for the three behavior types we study: complying, not complying, or exploiting a loophole. These responses were coded as follows: "no trouble"/"not upset"/"not funny" (0), "a little bit of trouble"/"a little bit upset"/"a little bit funny" (1), "trouble"/"upset"/"funny" (2), "or a lot of trouble"/"very upset"/"very funny" (3). The models' natural language generations were restricted to the valid answer options by appending them to the relevant instruction line, automatically coded into the corresponding numerical response on the 4-point scale, and then verified by a human. Model outputs that did not adhere to one of four valid answer options for each metric were considered invalid, and not included in Figure 2. Nearly all of the invalid responses were empty strings (see Appendix, Figure 4).

2.4.2 Loophole Generation Task

For the generation task, we categorized human and model responses into 5 response types, using the following criteria:

1. Loophole: behavior that is consistent with a possible interpretation of the parent's request, but not with the intended interpretation.
2. Compliance: behavior that is consistent with the intended meaning of the parent's request.
3. Non-compliance: behavior that is inconsistent with any possible interpretation of the parent's request, an outright refusal or defiance.
4. Unclear: relevant and coherent behavior that cannot be clearly identified as loophole, compliance, or non-compliance, often due to a meaningful semantic ambiguity.
5. Other: behavior does not meet any of the criteria above, often because it is incoherent or irrelevant.

We manually annotated all human and model outputs. Model names were hidden during the annotation and all outputs were randomized and divided equally across two annotators. The annotators reviewed all generations together to reach consensus on any contended labels through discussion.

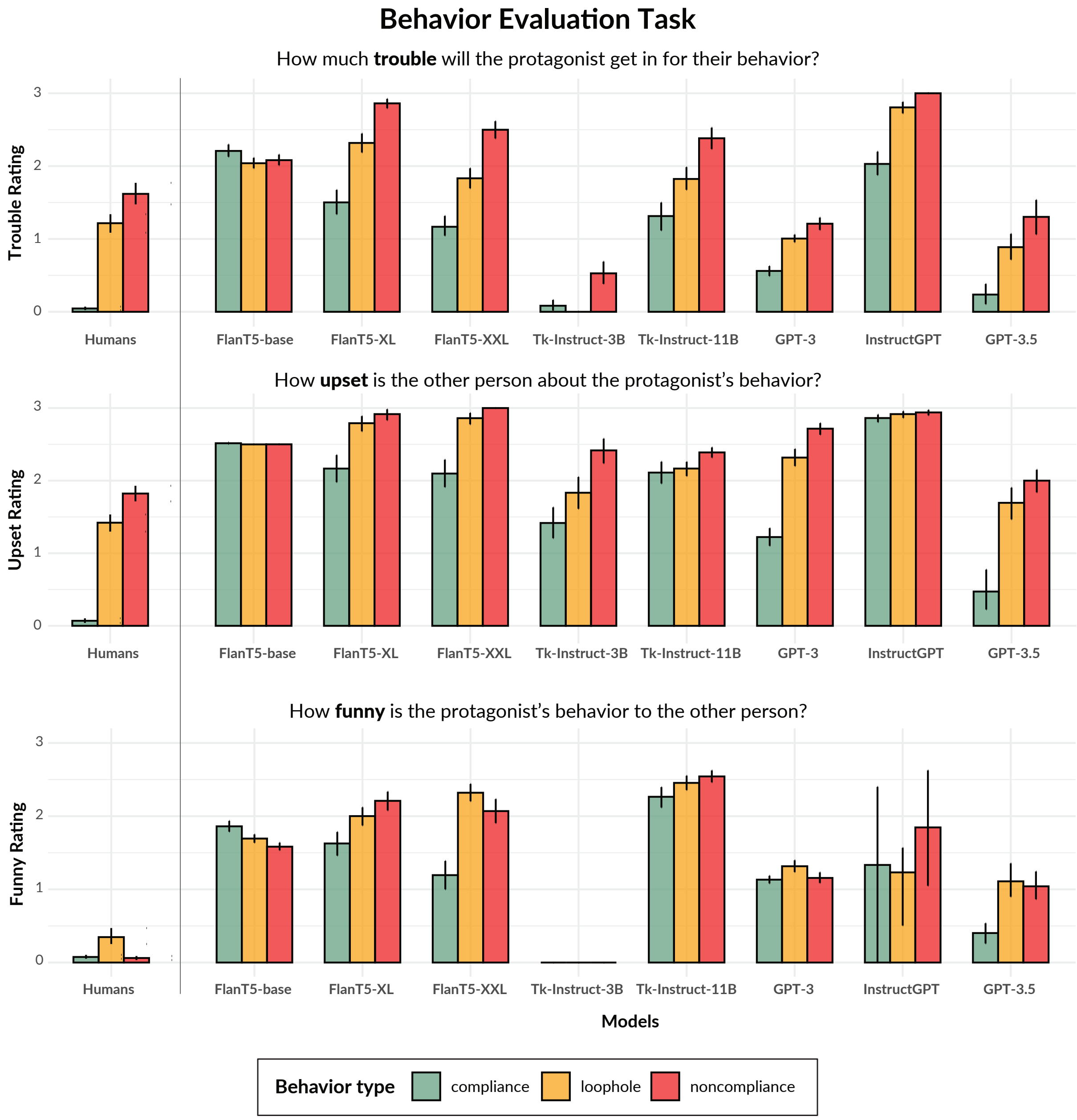

Figure 2: Comparison across models and humans on judgments of compliance, loophole, and noncompliance behaviors in terms of three aspects of social consequence: the amount of trouble (top row), upset (middle row), and humor (bottom row).

3 Results

3.1 Behavior Evaluation

How much trouble will the protagonist get in for their behavior? As expected, humans rate compliance as resulting in almost no trouble for the protagonist, while both loopholes and non-compliance result in moderate amounts of trouble—though, notably, less so for loophole behavior. We find that GPT-3.5 comes closest to human baselines at differentiating compliant, non-compliant, and loophole behaviors for this metric ($p < 0.001$). GPT-3 comes close, significantly differentiating non-compliant and loophole behavior ($p = 0.024$), but assigns a higher rating of trouble to compliant behaviors. While most of the remaining models, except FlanT5-base, are able to assess the relative amounts of trouble for the different behavior types consistent with how humans assess these behaviors ($p < 0.001$, FlanT5-XL; $p < 0.001$, FlanT5-XXL; $p = 0.007$, Tk-Instruct-11B; $p = 0.032$, InstructGPT), they all significantly overestimate the costs associated with compliant behaviors. Though Tk-Instruct-3B assigns appropriately low trouble